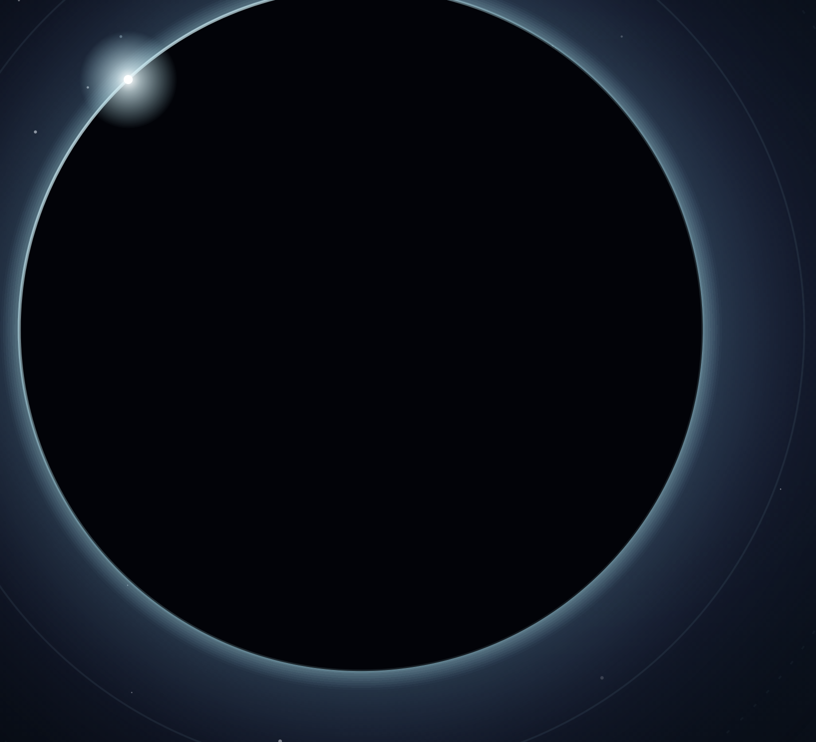
OBSERVATION LOG 03 · VIJAY SAI KILARI

BEYOND
THE KNOWN



What it costs to bring a large language model into existence, measured in **data**, **compute**, **energy**, and **time**. Some of it has been charted. Most of it is still dark.

FIELD · GENERATIVE AI INFRASTRUCTURE
OBJECTS · GPT · LLAMA · GEMINI · CLAUDE · GROK
EPOCH · 2017 → UNKNOWN

COST GROWTH	DATA	COMPUTE	ENERGY
2.4× / yr	**15.6T**	**16,384**	**310 GWh**
growth in frontier training cost since 2016	tokens used to pretrain Llama 3.1 405B	H100 GPUs running that single training run	estimated for Grok 4, about 28,700 US homes for a year

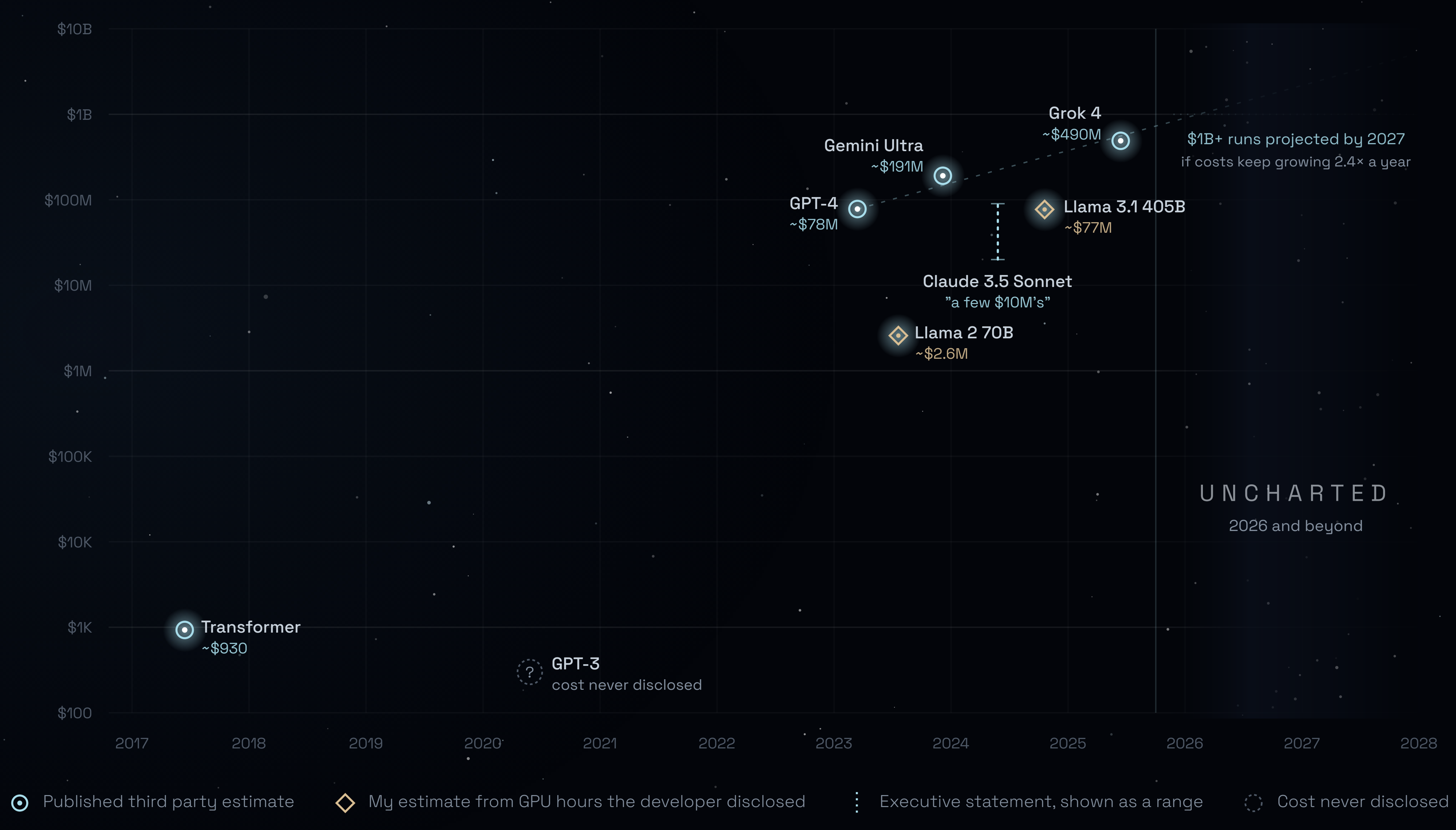
01 LAUNCH SEQUENCE

Eight stages take raw text to a deployed model. Nearly all of the compute, energy, and time is spent at stage four.

01	**Collect**	Web, books, code, licensed text
02	**Filter**	Remove duplicates, spam, personal data
03	**Tokenize**	Split text into word pieces
04 · CORE	**Pretrain**	Predict the next token, trillions of times
05	**Evaluate**	Benchmarks and checkpoints
06	**Fine tune**	Human written examples
07	**Align**	Human and AI feedback, red teaming
08	**Deploy**	Every request costs compute

02 STAR CHART OF KNOWN MODELS

Each point is a frontier model, placed by release year and estimated training cost. The scale is logarithmic: every gridline is ten times the one below. Past 2025, the map goes dark.



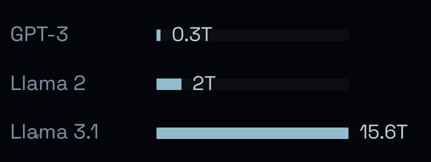
03 INSTRUMENT READINGS

The four resources that set the price of a model, and how far each has moved.

DATA

~50× in 4 years

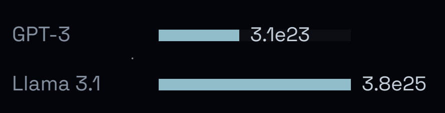
Training tokens, GPT-3 (2020) to Llama 3.1 (2024)



COMPUTE

~120× in 4 years

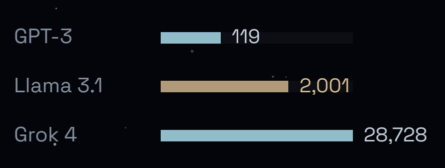
Floating point operations to train, log scale



ENERGY

119 → 28,728

US homes powered for a year by the training electricity, log scale. Sand marks my estimate



TIME

466 interruptions

in a 54 day stretch of Llama 3 pretraining, one every ~3 hours

54 days of pretraining

each line = one interruption day 1 → 54

04 SIGNAL STRENGTH

How much each developer has revealed about training. Only Meta transmits clearly; for the rest, we are reading faint signals.

Llama 3.1 405B	META · 2024	Tokens, GPU hours, hardware, emissions
GPT-3	OPENAI · 2020	Size, tokens, compute; cost withheld
GPT-4	OPENAI · 2023	Size, data, and hardware withheld
Claude 3.5 Sonnet	ANTHROPIC · 2024	Only "a few $10M's" from the CEO
Llama 2 70B	META · 2023	Tokens, GPU hours, emissions, dates
Grok	XAI · 2023 TO 2025	Grok 1 weights released; costs estimated
Gemini Ultra	GOOGLE · 2023	TPUv4 fleet named; cost estimated
Next frontier model	UNKNOWN · 2026+	Not yet observed

05 OPEN QUESTIONS

The numbers so far point toward a future no one has measured yet.

?01

When does the data run out?

Models may be trained on the entire stock of public human text sometime between 2026 and 2032.

?02

Who can afford the next one?

At 2.4× a year, the largest training runs pass $1 billion by 2027, leaving very few organizations able to build at the frontier.

?03

Where will the power come from?

Energy is only 2 to 6% of the cost, yet power and cooling now decide where clusters can be built at all.

?04

How much stays hidden?

Of seven frontier models charted here, only Meta's come with full training numbers. The rest are estimates.

WHAT WE DO KNOW Pretraining dominates the bill. Costs compound every year. And for most organizations, fine tuning an existing model costs a tiny fraction of training a new one.

Sources Brown et al. (2020) · Patterson et al. (2021) · Ouyang et al. (2022) · Bai et al. (2022) · Meta Llama 2 and 3.1 model cards · Llama Team (2024) · Gemini Team (2023) · Stanford AI Index (2024) · Cottier et al. (2024) · Villalobos et al. (2024) · Amodei (2025) · xAI (2024) · Tom's Hardware (2025) · Epoch AI (2025) · U.S. EIA. Estimates marked as mine multiply disclosed GPU hours by an assumed cloud price or by GPU power. Full references on the interactive page.

INTERACTIVE VERSION
omsairamkvs99.github.io/ml-portfolio